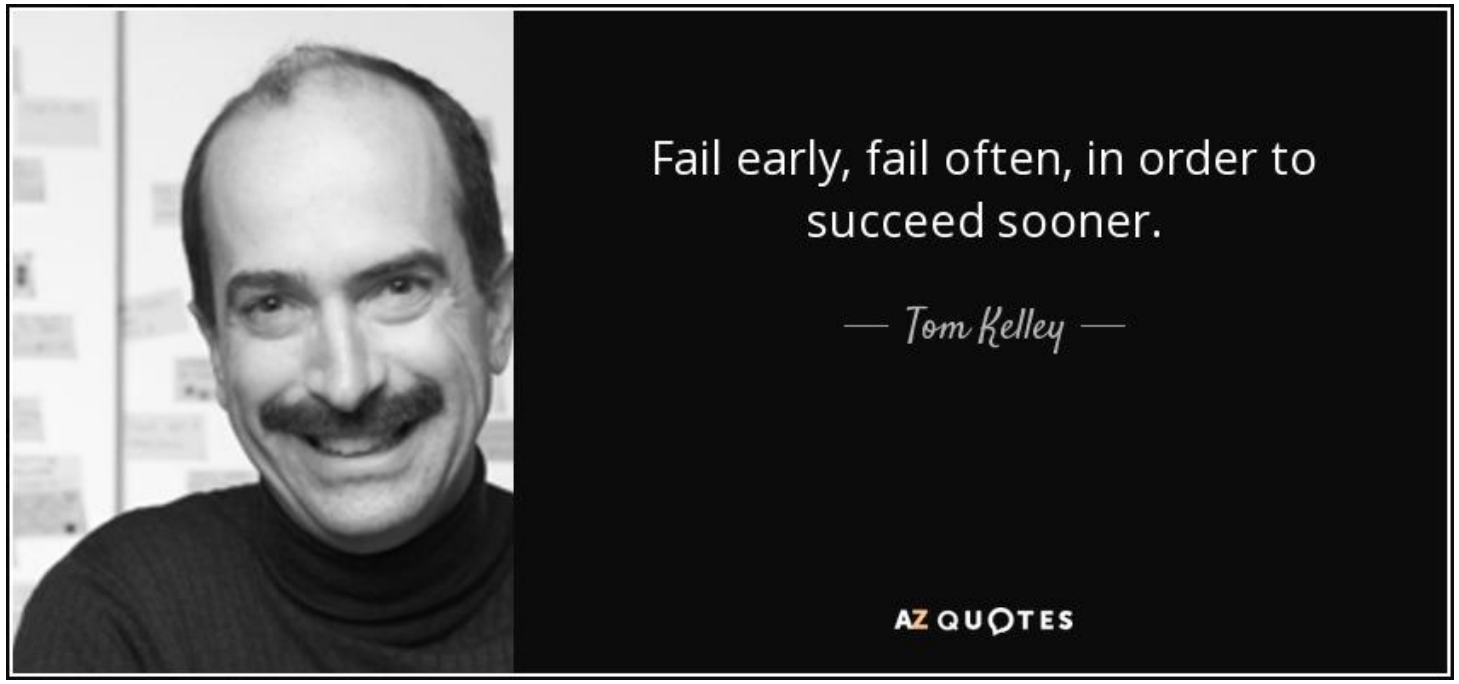
GROUP 4 CAPSTONE

By: Brandon A, Jason M, Keifer K, & William C

SECTIONS

- Prologue
- Planning
- Backend
- Full Stack
- Release
- Demo

A FALSE START



OUR INITIAL PLAN

Our initial plan was to each explore the codebase to find tasks. What made a task? Turns out none of us really knew.



ROLES AND DELEGATION

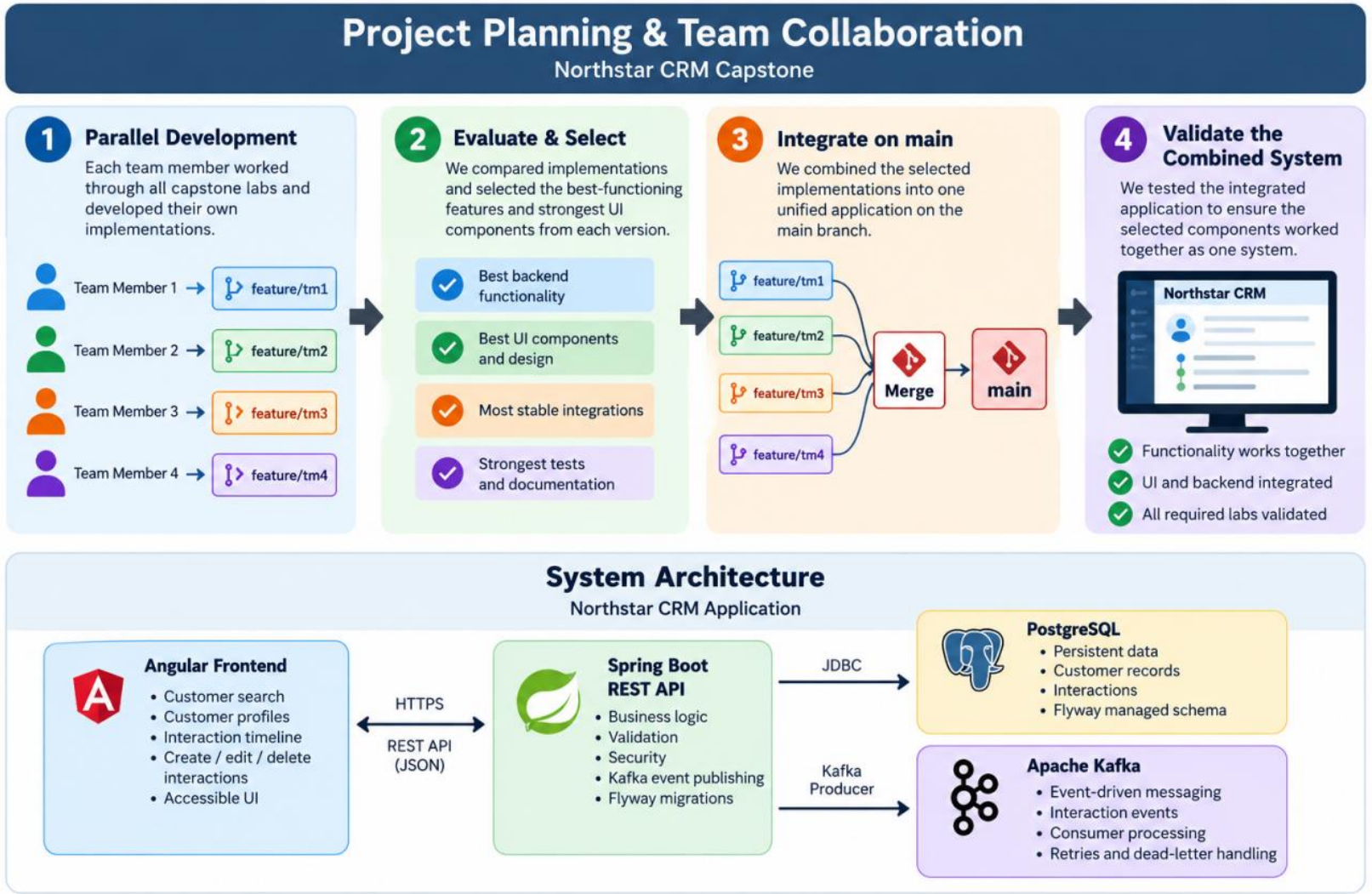
We went with an initial direction of working in a jack of all trades fashion. Each working on all aspects of the project.

However, as the project progressed, we found ourselves naturally owning larger sections of the codebase.

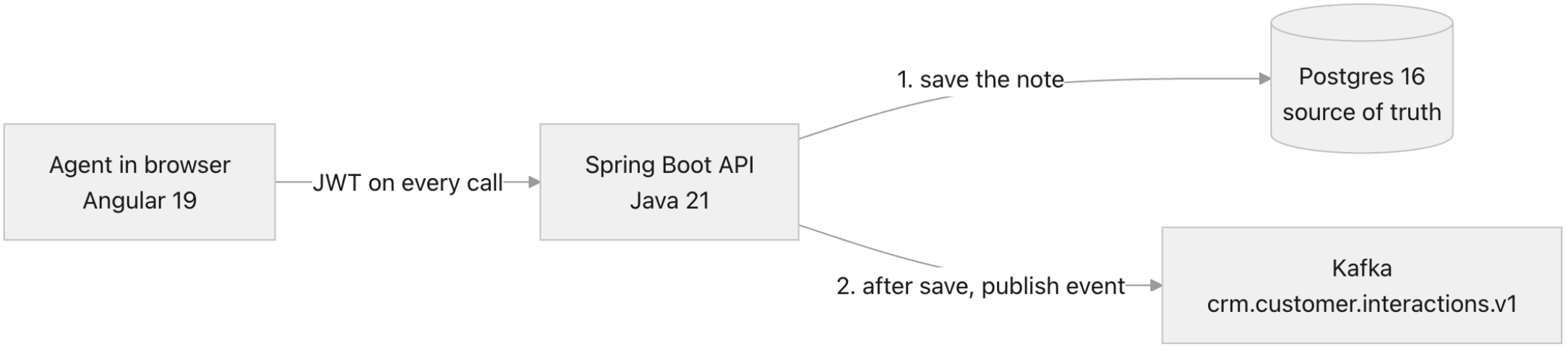
Keifer on the backend. Jason on the frontend. Brandon on database and review. Will on deployment and merging.



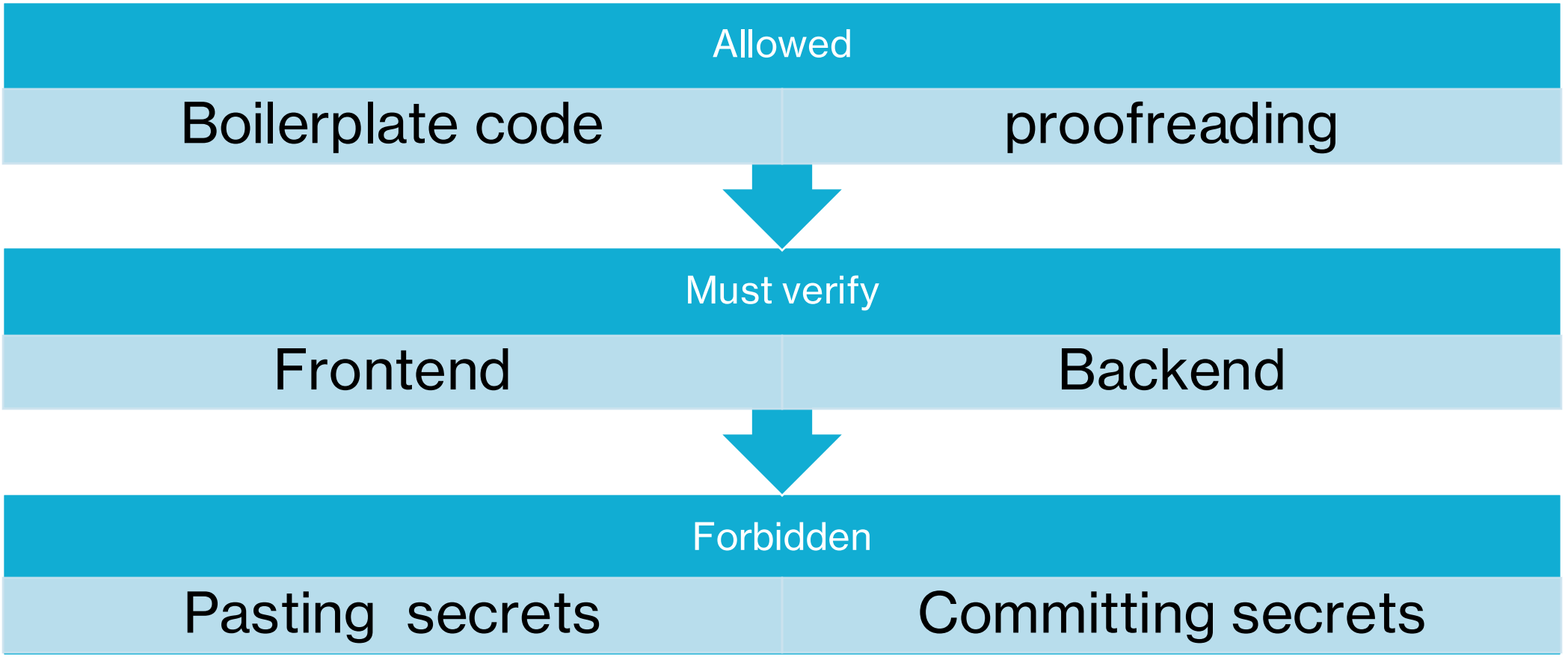
PLANNING


Project Planning & Team Collaboration
Northstar CRM Capstone
1 Parallel Development
Each team member worked through all capstone labs and developed their own implementations.
Team Member 1
feature/tm1
Team Member 2
feature/tm2
Team Member 3
feature/tm3
Team Member 4
feature/tm4
2 Evaluate & Select
We compared implementations and selected the best-functioning features and strongest UI components from each version.
Best backend functionality
Best UI components and design
Most stable integrations
Strongest tests and documentation
3 Integrate on main
We combined the selected implementations into one unified application on the main branch.
feature/tm1
feature/tm2
feature/tm3
feature/tm4
Merge
main
4 Validate the Combined System
We tested the integrated application to ensure the selected components worked together as one system.
Northstar CRM
Functionality works together
UI and backend integrated
All required labs validated
System Architecture
Northstar CRM Application
Angular Frontend
Customer search
Customer profiles
Interaction timeline
Create / edit / delete interactions
Accessible UI
HTTPS
REST API (JSON)
Spring Boot REST API
Business logic
Validation
Security
Kafka event publishing
Flyway migrations
JDBC
Kafka Producer
PostgreSQL
Persistent data
Customer records
Interactions
Flyway managed schema
Apache Kafka
Event-driven messaging
Interaction events
Consumer processing
Retries and dead-letter handling

ARCHITECTURE



AI USAGE PLAN



GITHUB



We enabled branch protections to ensure no errant merges or pushes to main would occur

BACKEND

REST APIs

- Strict Layering with DTO Isolation
- Jakarta Bean Validation at Boundary
- RFC 9457 & Correlation IDs

Security

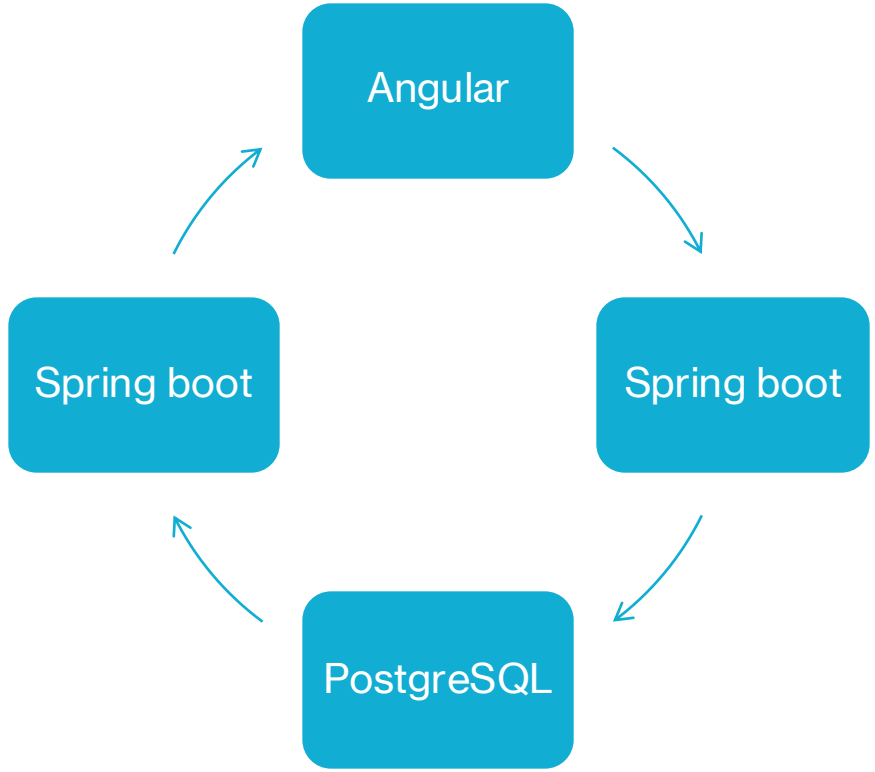
- Stateless JWT & RBAC
- Service Enforced Admin Guards
- JPA, Flyway, & Soft Deletes

Kafka Messaging

- Post Commit Publishing
- Idempotent Consumers
- DLT Error Routing
- Testcontainers Validation

FULL STACK

- Angular front end
  - Smooth agent journey
  - No hidden tasks or unusable buttons
- PostgreSQL database
  - Stores our customers and their interactions
  - Fast and responsive calls to and from
- Accessible UI
  - Clearly communicated UI
  - Redundant navigational tools



SECURITY AND AUTH



What was added:

- Login at POST /api/v1/auth/login.
    - Returns a signed 15-minute JWT
- All API calls require a token
    - Status code 401 is called when the user is not signed in.
- Two roles: admin and agent
    - Status code 403 is called for improper permissions
- The frontend keeps the token in memory to avoid the potential of script injection
- Agents *were* able to close a customer through an edit interaction, despite close being an admin only permission. Another layer of verification was added to this interaction to prevent permission leaks.

CI/CD - THE 3 GATES



RELEASE

After CI/CD passes cleanly, the JAR and frontend build are downloaded from the previously run tests, no rebuild needed.

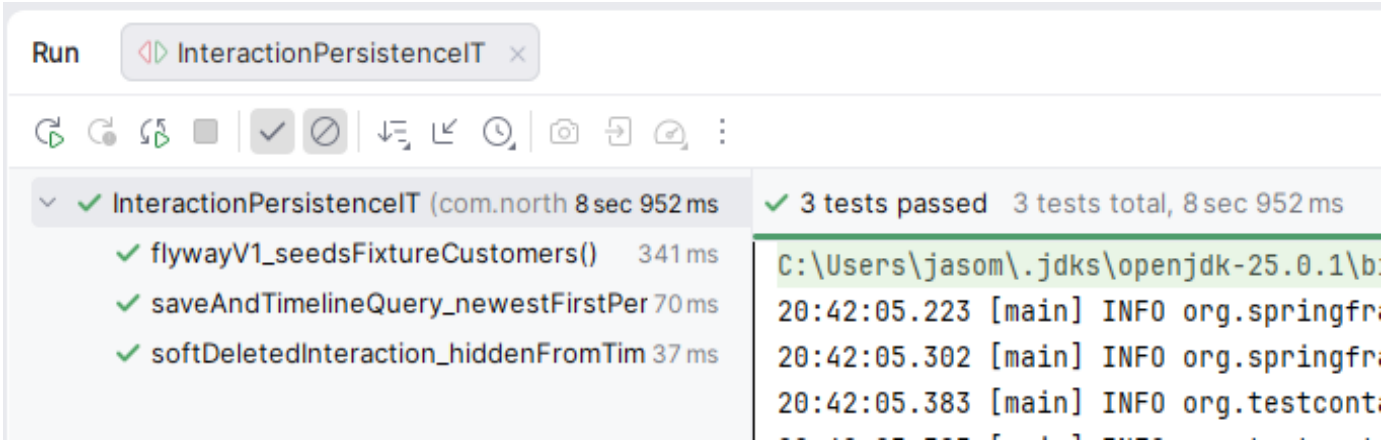
Both images' digests are recorded on the GitHub container repository, and a SHA-256 checksum is generated for this release.

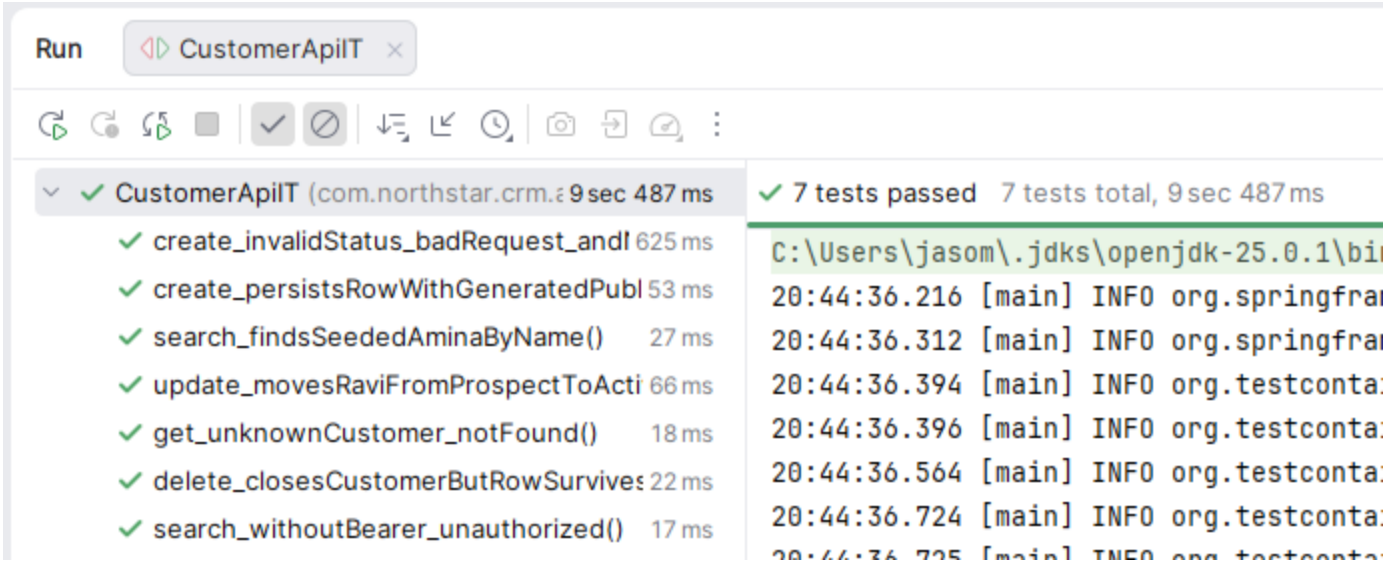
We were not able to get our project deployed to Kubernetes in the time we had.

DEMO RECORDING

Group 4 Capstone Demo

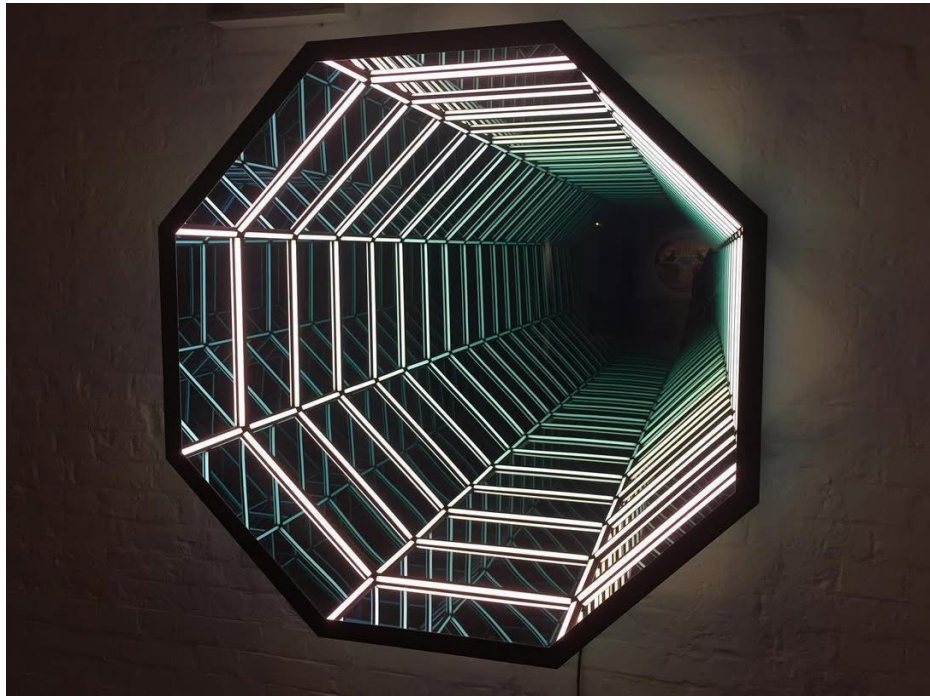
EVIDENCE


Run
InteractionPersistenceIT
InteractionPersistenceIT (com.north 8 sec 952 ms
flywayV1_seedsFixtureCustomers() 341 ms
saveAndTimelineQuery_newestFirstPer 70 ms
softDeletedInteraction_hiddenFromTim 37 ms
3 tests passed 3 tests total, 8 sec 952 ms
C:\Users\jasom\.jdks\openjdk-25.0.1\b
20:42:05.223 [main] INFO org.springfr
20:42:05.302 [main] INFO org.springfr
20:42:05.383 [main] INFO org.testcont


Run
CustomerApiIT
CustomerApiIT (com.northstar.crm.a 9 sec 487 ms
create_invalidStatus_badRequest_andI 625 ms
create_persistsRowWithGeneratedPubl 53 ms
search_findsSeededAminaByName() 27 ms
update_movesRaviFromProspectToActi 66 ms
get_unknownCustomer_notFound() 18 ms
delete_closesCustomerButRowSurvives 22 ms
search_withoutBearer_unauthorized() 17 ms
7 tests passed 7 tests total, 9 sec 487 ms
C:\Users\jasom\.jdks\openjdk-25.0.1\bi
20:44:36.216 [main] INFO org.springfra
20:44:36.312 [main] INFO org.springfra
20:44:36.394 [main] INFO org.testconta
20:44:36.396 [main] INFO org.testconta
20:44:36.564 [main] INFO org.testconta
20:44:36.724 [main] INFO org.testconta

RETROSPECTIVE

The project could have been smoother in the beginning, but we were happy with our collective ability to deliver an adequate solution to the problems we were given. We were also excited to expand our initial capabilities by doing some fun/extra work. With this project, we respectively developed our skills in the various materials we covered through practice. This practice should foster our knowledge going forward given that experience is the greatest way to learn.



ABOVE AND BEYOND (CHANGE TITLE)

- We made it accessible
- Progressive UI
- Confetti.js



THANK YOU!

